

Rigging & Slinging

An equipment register, pre-use inspections and sling load calculations

⚠ FOR UNDERSTANDING PURPOSES ONLY

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Flutter / Dart

Equipment library

Inspection

Calculation

PDF & Excel

1. What It Is

Rigging equipment such as wire rope slings, web slings, chain slings and shackles must be tracked, inspected and used within its limits. This large screen — the biggest in the app — does all three in **three tabs**. It uses the same dark palette and collapsible numbered sections as the Crane Lifting Plan. It is reached from Safety › Process Safety › Lifting Operations › Rigging and Slinging.

Tab	Purpose
Equipment Library	The register of every item, with status and next-due dates. Add, edit, delete and export to Excel
Inspection	A pre-use checklist for the selected item, saved with its history and exported as a PDF
Calculation	Working load and factor-of-safety checks, and sling tension for different hitches

Equipment records and inspections are kept in the app database, so they survive restarts.

2. What Is Recorded for Each Item

Category	Wire Rope Sling, Web Sling, Chain Sling, Shackle, or a category you create
Equipment ID	Your own unique ID, such as WRS-001
SWL / WLL	Safe working load in tonnes
Description	Free text
Last inspection	A date, or a dash if never inspected
Status	Pending (never inspected), Pass or Fail

3. Tab 1 — Equipment Library

Two sub-pages switch with a blue strip.

Equipment List

- **Summary:** total items, inspected items and overdue items, which are highlighted in red.
- **Register table** that scrolls sideways: serial number, category, ID, working load, description, status, last inspection and next due date.
- **Colour coding:** grey for Pending, green for Pass, red for Fail and an orange cell for an overdue date.
- **Row menu:** view details, load the item into the Inspection tab, or jump to edit it. Each row also has a small PDF button that exports that item's latest saved inspection.
- **Export Excel:** writes the whole register to a spreadsheet and shares it.

The **next due date** is worked out when the screen loads, from each item's last inspection and the frequency saved with it (for example 3 months).

Manage Equipment

Section	What it does
Add New Equipment	Equipment ID, working load, description and category
Update Category	Change the category of an existing item without opening the full edit form
Edit Equipment	Search for an item and unlock its fields to change the ID, load, description, category or certificate date. Deleting asks for confirmation

4. Tab 2 — Inspection

Three collapsible sections, with Save Inspection, Export PDF and a fold-all toggle at the top.

Section	Content
A. Equipment details	Search for an item. Its ID, category, working load, last inspection and status appear as a read-only card
B. Checklist	The checklist for the item's category, grouped into subsections. Each item cycles on a tap. A remarks box opens for items that need comments
C. Sign-off & outcome	The overall outcome (Pass or Fail) is worked out automatically. Then the competent rigger, the safety officer, the inspection date and the next inspection frequency in months or years

How the checklist behaves

- Items are either **critical** (marked ★ below) or minor.

- **Minor items** cycle N/A → OK → Caution → Fail. Caution notes a small issue.
- **Critical items** cycle N/A → OK → a problem state. Any problem on a critical item fails the sling.
- **The overall result is automatic.** A critical problem or any failed minor item makes the outcome **FAIL**, and it cannot be overridden. Otherwise, once something has been checked and nothing failed, it is **PASS**.
- Items marked not applicable are left out of the result.

The four checklists

Category	Subsections	Examples of what is checked
Wire rope sling (7)	Identification · General condition · Diameter and wear · Structural integrity · End termination · Lubrication · Usage condition	Tag and working load marked★, no broken wires★, bird-caging★, kinks★, crushing★, heat damage★, flat spots★, core protrusion★, ferrule intact★, shock-load damage★
Web sling (6)	Identification · General condition · Stitching · Eye section · Usage condition · Hardware	No cuts or tears★, chemical or heat damage★, stitching intact★, no broken threads★, no knots★, hooks OK★
Chain sling (8)	Identification · Chain links · Wear and thickness · Master link · Hooks · Connecting components · Assembly · Usage	Tag, working load and grade marked★, no cracks★, bent links★, elongation above 5%★, wear above 10%★, safety latch works★, no missing parts★
Shackle (8)	Identification · Body · Pin · Thread and securing · Alignment and fit · Wear and distortion · Usage · Compatibility	Working load marked★, no cracks★, original pin★, pin secured★, no side loading★, load in the bow not on the pin★, correct size for the load★

A custom category shows a “coming soon” note instead of a checklist.

Saving and the PDF

1. Saving checks that an item is selected.
2. The item’s status and last-inspection date are updated in the register.
3. The full inspection is stored: every state, every remark, the outcome, the dates, the people and the frequency.
4. A message confirms the result, for example “saved as PASS”.

The PDF has a title bar (green for Pass, red for Fail), the equipment details, the full checklist with remarks under any flagged items, a summary of counts and a personnel strip.

5. Tab 3 — Calculation

Three numbered sections. Inputs are pale yellow and results are dark read-only cells. Unit toggles switch between metres and feet, and tonnes and kilograms.

01 — Wire sling working load and factor of safety

Part	How it works
Estimated working load	A rule of thumb: diameter in inches squared $\times$ 8 gives tonnes. Millimetre entries are converted to inches first
Actual working load	The certificate value, if you have it, which overrides the estimate
Factor of safety	Minimum breaking load $\div$ working load, shown as a number and a ratio such as 4 : 1
Alarm	Green “Safe to lift” if the factor is 1 or more; red “Danger — do not lift” if it is below 1

Rule of thumb only

The estimate is approximate. The true working load must always come from the manufacturer and its certificate.

02 — Sling load: tension and angle

Enter the sling height H, sling length L, load weight W and the sling’s working load (filled in from section 01). Choose a hitch to see the tension in each sling and its effective capacity.

Hitch	Factor	Result
Single vertical	1.0	Tension = load. Capacity = working load
Choker	0.75, or adjusted by an angle slider from 30° to 180°	Tension = load $\div$ factor
Basket, straight	2.0	Tension = load $\div$ 2. Capacity = 2 $\times$ working load
Basket, angled	Depends on H and L	Tension = load $\times$ (L $\div$ H) $\div$ 2
2-leg bridle	2 $\times$ cosine of the angle	Tension = load $\times$ (L $\div$ H) $\div$ 2
4-leg bridle	3 $\times$ cosine of the angle	Tension = load $\times$ (L $\div$ H) $\div$ 3, since in practice three legs carry the load

The sling angle is colour-coded: **60° or more green, 45° or more orange, below 45° red (unsafe)**. A warning also appears when the angle is steep. The result line reads “Safe to lift” when the tension is within the effective capacity, and “Do not lift” otherwise. Hitch pictures are shown for vertical, choker, basket and angled basket.

03 — Lug calculation

Two sub-tabs, **Load Calculation** and **Lug Analysis**. Both are placeholders marked “coming soon”. Lug-plate tear-out, weld capacity and dimension analysis are planned for a future update.

Exporting the calculation

A reference name, date and optional remarks can be added, and the calculations exported as a PDF summary.

6. How It Fits Together

One item, three views

An item is registered in the Library, inspected in the Inspection tab (which updates its status and next due date in the Library), and its working load feeds the Calculation tab. The Excel and PDF exports give a paper trail for audits.